

| Question | Answer                                                                                                                                                                                           | Marks     |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 2(a)     | <ul style="list-style-type: none"> <li>• resistance of a metal</li> <li>• volume of a gas at constant pressure</li> <li>• e.m.f. of a thermocouple</li> </ul> <i>Any two points, 1 mark each</i> | <b>B2</b> |
| 2(b)(i)  | $Q = mc\Delta T$                                                                                                                                                                                 | <b>C1</b> |
|          | evidence of realisation that $Q$ lost by water = $Q$ gained by mercury                                                                                                                           | <b>C1</b> |
|          | $18.7 \times 4.18 \times (37.4 - T) = 6.94 \times 0.140 \times (T - 23.0)$                                                                                                                       | <b>C1</b> |
|          | $T = 37.2\text{ }^{\circ}\text{C}$                                                                                                                                                               | <b>A1</b> |
| 2(b)(ii) | use a liquid with a lower (specific) heat capacity (than mercury)<br><b>or</b><br>use a smaller mass of mercury                                                                                  | <b>B1</b> |
| 2(c)(i)  | depends on properties of a real substance                                                                                                                                                        | <b>B1</b> |
|          | $0\text{ }^{\circ}\text{C}$ is not absolute zero                                                                                                                                                 | <b>B1</b> |
| 2(c)(ii) | ideal gas                                                                                                                                                                                        | <b>B1</b> |

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|----------|--------|-------|
| 2(c)     | 2      | M1    |